

Review Article

Fraud Prediction and Prevention in Mobile Money Payment Systems (MMPS): A Systematic Literature Review of Text-Based Detection Methods

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Mobile money services have become essential in providing financial inclusion, particularly in underserved and developing regions, by enabling access to banking, payments, and commerce through mobile devices. Despite their benefits, these services are increasingly targeted by fraudsters employing sophisticated text-based schemes such as phishing, spoofing, and fraudulent short message services (SMS) verification codes. This review of the literature examines how to predict and stop text-based fraud in mobile payment systems. In this review, we systematically analyzed 160 studies from eight databases between 2017 and 2023 to measure, map, and investigate papers released during this period. Research gaps in mobile money payment systems (MMPS) have been highlighted, along with prevalent fraud types that occur in these systems. There was not a single paper that used the NLP approach to identify fraud in MMPS.

Keywords: fraud prediction; fraud prevention; machine learning; mobile money; mobile money transfer; text-based fraud

1. Introduction

Mobile money represents a transformative innovation within financial technology (fintech), enabling users to conduct monetary transfers and savings transactions using mobile phones without the necessity of traditional banking infrastructure [1]. Unlike mobile banking, mobile money services—such as EcoCash, M-Pesa, and Orange Money—are not inherently tied to conventional bank accounts or credit cards [2]. Instead, mobile money systems operate through basic telecommunications protocols, including unstructured supplementary service data (USSD) codes and short message service (SMS), thus facilitating access even via low-end feature phones and circumventing the need for internet connectivity [3].

These platforms extend financial services to previously excluded populations by enabling peer-to-peer payments, retail transactions, savings, credit, and remittance services [4]. Consequently, mobile money is widely acknowledged as

a critical mechanism for fostering financial inclusion and poverty alleviation across developing economies [5]. Its proliferation has dramatically enhanced access to financial services for marginalized populations, encouraging domestic savings, promoting entrepreneurship, and stimulating microeconomic activity [6, 7]. In Sub-Saharan Africa, for instance, over 90% of telephone lines are mobile, creating an enabling environment for mobile money platforms to thrive amidst traditional financial institutions concentrated primarily in urban centers [8, 9].

Despite its transformative potential, mobile money is not devoid of risks and vulnerabilities. A significant proportion of mobile money providers lack comprehensive privacy policies or fail to offer them in languages accessible to the majority of their client base [5]. The leakage of sensitive information, particularly SMS-based verification codes, has emerged as a notable security threat [3]. Furthermore, mobile money users often possess limited financial and

digital literacy, rendering them susceptible to diverse forms of fraud, particularly social engineering attacks [7].

Enhancing transaction security is, therefore, paramount. Machine learning (ML) classifiers have demonstrated considerable promise in predicting mobile money transfer fraud by detecting anomalous transaction patterns [1, 2]. Prevailing mitigation strategies advocate for standardizing transactional procedures, instituting joint action mechanisms among stakeholders, and enforcing stringent regulatory frameworks [4, 6]. Nevertheless, fraudulent call-based transactions remain a pervasive challenge, disproportionately impacting consumers and telecom service providers across developing nations [9]. Proposed interventions include improving fraud reporting channels, enhancing user education initiatives, and systematically addressing vulnerabilities within the mobile banking and payment ecosystems [5].

Innovations in authentication, such as voice verification techniques including speech biometrics and speech recognition, are increasingly integrated into mobile money services to bolster security during transactions [7]. These technologies are capable of augmenting existing payment infrastructures by providing more robust mechanisms for user verification and fraud prevention.

While contemporary research has predominantly concentrated on leveraging ML models, rule-based systems, and biometric techniques to detect and mitigate mobile money fraud, the exploitation of linguistic vulnerabilities, specifically within text-based communication, remains underexplored. Notably, fraudulent practices frequently exploit SMS messaging and social engineering, domains fundamentally anchored in language manipulation.

Natural language processing (NLP), a subfield of artificial intelligence dedicated to interpreting human language, offers considerable potential to bridge this gap. Techniques such as sentiment analysis, named entity recognition, topic modeling, and text classification provide powerful tools for detecting deceptive communications and anomalous user inputs. Although NLP methodologies have demonstrated significant efficacy in domains such as email phishing detection and online scam identification, their application within mobile money fraud detection remains remarkably limited [10, 11].

Integrating NLP into fraud detection frameworks introduces a paradigm shift by enabling systems to detect fraud vectors that exploit linguistic features such as deceptive SMS messages, phishing attempts, and manipulated communication flows, alongside conventional numerical transaction analysis [10, 12]. Traditional ML models, while proficient at identifying statistical anomalies in transactional data, often lack the semantic sensitivity required to discern fraud embedded within textual interactions.

Emerging research underscores this potential. For example, a study published in ML demonstrated that incorporating text-sequence features into fraud detection models for online and mobile banking substantially improved classification performance relative to traditional models, without necessitating extensive feature engineering [11]. Similarly, a comprehensive review in the *Journal of Big Data* emphasized the critical role of real-time data processing frameworks, augmented by NLP techniques, in

enhancing credit card fraud detection systems by uncovering complex linguistic patterns indicative of fraud [13].

Despite such promising findings, the domain remains underdeveloped in the context of mobile money ecosystems. There is a dearth of systematic research exploring how textual data—transaction descriptions, SMS alerts, and customer communications—can be harnessed through NLP to enhance fraud detection within mobile money payment systems (MMPS) environments. This lacuna calls for rigorous investigation into NLP-augmented models tailored for mobile money fraud prevention.

Addressing this gap is not merely a technical imperative but a socioeconomic necessity. Financial losses borne by customers due to mobile money fraud have far-reaching implications, undermining trust in digital financial systems, exacerbating inequalities, and impeding broader efforts toward financial inclusion [2, 5]. Psychological distress, loss of social cohesion, and diminished faith in regulatory and legal institutions are additional adverse effects documented in the aftermath of fraud incidents [3].

Consequently, there is an urgent need for holistic strategies that combine technological innovation, particularly the integration of NLP, with user education, community-based awareness initiatives, and the reinforcement of regulatory frameworks. A multidisciplinary approach is imperative to ensure that mobile money continues to be a vehicle for financial empowerment rather than a conduit for financial victimization.

In conclusion, while traditional fraud detection methodologies retain considerable merit, the integration of NLP-based techniques presents a promising frontier for enhancing the robustness and comprehensiveness of fraud detection in mobile money systems. Future research must systematically explore and empirically validate NLP's effectiveness within this domain to foster the development of resilient, intelligent, and contextually adaptive fraud prevention frameworks.

2. Related Literature

Research on SMS fraud prediction and prevention within MMPS has garnered increasing scholarly attention due to the proliferation of cyber-enabled financial crimes in low- and middle-income economies [14]. Empirical studies demonstrate that conventional ML algorithms, such as logistic regression, random forest, and support vector machines (SVM), can identify suspicious user profiles and predict fraudulent transactions with a reasonable degree of accuracy. Nevertheless, the escalating sophistication of fraud schemes necessitates continual methodological innovation to maintain resilience against evolving threats [15].

From the corpus of 160 studies included in this review, only a small fraction (0.8%) explicitly focused on SMS-based fraud in MMPS [16]. Within this subset, anomaly detection was primarily approached through artificial intelligence and ML models, notably deep learning architectures, decision trees, SVM, and logistic regression. While these methods have demonstrated value, their scope is largely constrained to numerical transaction features and metadata patterns (as indicated in Tables 1 and 2) [17–20].

TABLE 1: Measures for writing systematic literature review questions [20].

Criteria	Details
Population	Mobile Money Users, Security Experts, Mobile Money Service Providers, Fraud Prevention Agencies and Regulatory Bodies
Intervention	Methods, algorithms, and techniques utilized in the detection and prevention of SMS-based fraud in mobile money transactions
Outcome	Development of an NLP-Enhanced Security System, Efficient and User-Centric Verification System, Adaptable System for Evolving Threats
Context	Prediction and prevention of mobile money fraud committed through SMS/text

Note: Source: Authors' work.

TABLE 2: Publications were initially sourced from six databases consulted.

Databases	URL	First total	Initial filtering	Last filtering
MDPI	https://www.mdpi.com/	59	47	12
Elsevier	https://www.sciencedirect.com/	73	42	31
Emerald Insight	https://www.emerald.com/	39	22	17
IEEE Xplore	https://www.ieee.org/	75	43	32
ACM Library	https://dl.acm.org/	25	10	15
Wiley Online	https://onlinelibrary.wiley.com/	35	22	13
Springer	https://www.springer.com	85	72	13
Taylor & Francis	https://www.tandfonline.com/	44	42	2
Google Scholar		222	197	25
Total		657	497	160

Note: Source: authors' work.

Notably, there is a significant absence of systematic integration of NLP for analyzing the textual content of fraud-related communications. This gap persists despite emerging evidence from adjacent domains such as online banking and credit card fraud detection; that NLP-based features, when combined with transaction data, can outperform conventional classifiers while preserving data privacy [21]. For example, BERT-based smishing detection models and hybrid approaches incorporating regular expression analysis have achieved high accuracy in identifying malicious messages that blend languages (e.g., Arabic–English hybrids). Likewise, smishing detection applications employing Naïve Bayes (NB) or deep-learning hybrids have reported accuracy rates exceeding 99%, underscoring the potential of language-aware fraud detection strategies [22].

Beyond individual model performance, recent literature also emphasizes the need for adaptive, feedback-driven systems that can learn from new fraud patterns in real time. The feedback loop paradigm (Figure 1) whereby models iteratively retrain on fresh instances of fraudulent and legitimate activity addresses the static nature of conventional fraud detection systems and strengthens resilience against adversarial adaptation.

2.1. Research Gap. While ML has been extensively applied to MMPS fraud detection, current approaches disproportionately emphasize structured, nonlinguistic data (e.g., transaction amounts, timestamps, and device IDs) and neglect the linguistic dimension of fraud, despite the centrality of SMS and other text-based channels in the operational tactics of fraudsters. Keyword trends from the systematic review, highlighting terms such as smishing, phishing, BERT, and tokenization, suggest that the research community recognizes

the relevance of NLP but has yet to institutionalize its application in MMPS fraud detection [23].

Furthermore, few studies have explored the fusion of NLP with adaptive learning strategies in the MMPS context, despite strong indications from adjacent sectors that such integration could enhance detection accuracy, adaptability, and explainability. The absence of comprehensive, context-specific frameworks combining textual analysis with transaction-based anomaly detection leaves a critical methodological and practical void—particularly in regions where MMPS constitutes the dominant mode of financial exchange. Figure 1 demonstrates the linkage between NLP and adaptive learning systems [24].

2.2. Research Questions. Building on the identified need for dynamic, adaptive, and linguistically informed fraud detection systems (Figure 1), this review is guided by the following research questions:

RQ1. What are the common fraud types in MMPS?

RQ2. What adaptive learning strategies have been proposed or implemented to enhance fraud detection resilience in mobile money platforms?

RQ3. What methods have been employed to detect fraud in MMPS, and how do they compare in terms of performance and applicability?

2.2.1. Integrating NLP Framework Into Mobile Money Fraud Detection. MMPS have revolutionized financial inclusion in developing economies, but a surge in fraud has accompanied this growth. Traditionally, MMPS fraud defenses have relied on static rules and simple statistical checks. However, rule-

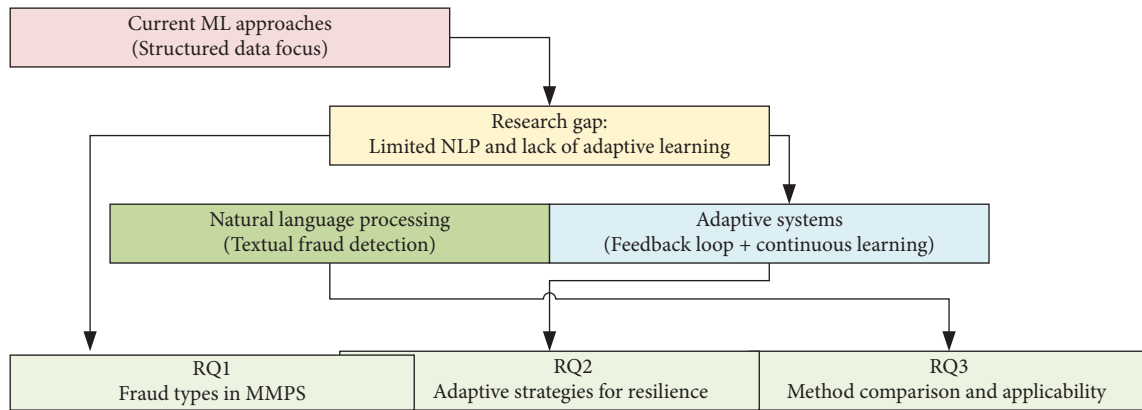


FIGURE 1: Conceptual diagram of NLP and adaptive systems.

based systems in mobile money have failed to stem the tide of fraud, as evidenced by continual reports of new fraud cases [17]. The dynamic and clever tactics of fraudsters outpace hard-coded rules, creating an urgent need for adaptive, intelligent detection methods. However, NLP offers a promising frontier for fraud detection by leveraging semistructured and unstructured textual data pervasive in MMPS, as shown in Figure 1.

The proposed MMPS fraud detection framework consists of four critical, sequential stages designed to enable real time, and explainable detection of fraudulent SMS-based transactions. This framework integrates NLP and ML within a secure and scalable architecture to detect, interpret, and respond to potential fraud incidents in mobile money ecosystems.

Step 1: The fraud detection lifecycle commences when a malicious actor, often leveraging social engineering techniques—crafts and disseminates phishing messages via SMS. These deceptive messages, typically designed to mimic legitimate institutions, are sent to unsuspecting users through mobile applications or embedded links. Once a victim engages with the fraudulent prompt, the interaction initiates a transaction request that is routed through a mobile money payment gateway. This step encapsulates the initial exploitation phase, where adversaries attempt to deceive the user into authorizing unintended financial transactions.

Step 2: Following the initiation of the transaction, the request is intercepted by an NLP-based fraud detection module, which serves as a secure, intermediary analytical platform. This module utilizes advanced language models (e.g., Transformer-based architectures such as BERT) to scrutinize the linguistic and contextual characteristics of the SMS content. By referencing historical data, such as past transaction records, behavioral usage patterns, and metadata extracted from system logs, the model determines whether the transaction is legitimate or indicative of fraudulent intent. Features, including the presence of phishing keywords, suspicious URLs, or anomalous message syntax, contribute to a probabilistic risk classification output.

Step 3: ML Fraud Scoring Engine determination. Once the NLP component classifies the transaction, the result is fed into a dedicated ML Fraud Scoring Engine. This engine evaluates the transaction further using a hybrid ensemble model. If the transaction is adjudged legitimate based on the computed fraud risk score, it is allowed to proceed, and the funds are transferred to the recipient. Conversely, if the transaction is flagged as suspicious, it undergoes an additional verification process. This may include multifactor authentication, manual review by a fraud analyst, or cross-validation with centralized fraud intelligence databases. If, upon verification, the sender is deemed authentic, the transaction is permitted; otherwise, it is blocked, and an incident report is generated for auditing and forensic tracking.

Step 4: In cases where the transaction is confirmed as fraudulent, the framework initiates a postincident response protocol. This includes the immediate reversal of the transaction (where feasible), notification to affected parties, and secure logging of the incident for regulatory compliance. Importantly, the fraud detection system is updated with the newly confirmed fraudulent transaction data, thereby strengthening future prediction accuracy. This dynamic feedback loop supports continuous learning and adaptive model refinement, ensuring that the system evolves in response to emerging fraud patterns and adversarial tactics. Fraud patterns in MMPS are constantly evolving; fraudsters devise new tactics once old ones are detected. A fraud detection framework must learn and adapt from each confirmed case to maintain efficacy. Our proposed framework incorporates a feedback loop that uses outcomes of investigations and confirmed fraud incidents to update and refine the models over time. Data augmentation and labeling ensure that a confirmed fraudulent transaction is fed back into the training data, where the NLP models and the integrated fraud scoring model are fine-tuned and retrained with the new labeled data periodically. Adaptive thresholds and rules ensure that if the NLP model's output consistently flags a certain type of phrase that turns out to

correspond to fraud, the system can adopt a more aggressive stance for that pattern. Analyst feedback integration ensures fraud alerts are reviewed and provides feedback. Our framework captures this expert feedback as well. Deployment of new detectors has an extensible architecture that ensures that if a novel data source or a new NLP technique becomes relevant, the feedback process might indicate a need for it.

Through the feedback loop, the framework achieves a learning ecosystem; the more fraud and nonfraud it encounters, the smarter it gets. The static nature of the present fraud systems is one of their main drawbacks, which is immediately addressed by this cycle of ongoing development. The feedback loop ensures that the integration of NLP is sustainable and resilient against the adaptive strategies of fraudsters by addressing the static nature of the current fraud systems.

This review aimed to investigate the effectiveness of MMPS deployed to deliver financial services in the broader mobile money context. Accordingly, this systematic literature review was guided by research questions explicitly formulated to address these concerns. The review questions were motivated by the imperative.

3. Materials and Methods

A comprehensive review of the literature should always be conducted using an unbiased approach to ensure consistency when analyzing all relevant studies on the specified issue. In this case, the authors in [26–28] were consulted to guide the three essential steps of conducting a stand-alone SLR, which include “the review plan, how to carry out the review, and the review reporting process.” These steps present a thorough and consistent procedure for conducting a systematic literature review in this study. Although the concentration of this analysis is on SMS/text-based fraud prediction and prevention in MMPS, the study provides extensive insights that can be valuable to academics researching any mobile money platform. For a comprehensive overview of Okoli’s recommendations for a systematic literature review methodology, see Figure 2.

3.1. Planning the Review. To characterize the most recent 7 years’ worth of articles on student learning performance, a systematic review was carried out. Planning is required to start any review process, according to [20]. As a result, it is imperative to decide on and create a meticulous plan for the review procedure.

3.2. Conducting the Review. Building digital library search strings, obtaining studies from databases, choosing and honing studies, extracting data, and delivering a final analysis are the five main processes in the literature search process. To find pertinent material, the search terms are as follows:

- (“emerging threats OR attacks OR fraud OR challenges”) AND (“mobile money transactions OR services OR m-payment”),

- (“fraud” OR “mobile money” OR “mobile wallet” OR transfer OR “m-payment”),
- (“NLP approaches” OR “techniques” OR “Methods” AND “detecting fraud” OR “threats” OR “attack”) AND (“mobile money transactions” OR “Services” OR “M-payment”),
- (“algorithms OR techniques”) AND (“detecting OR predicting”) AND (“sms-based fraud” OR “threats OR challenges”) AND (“mobile money transaction” OR “m-payment”),
- “Existing mitigation strategies” AND “mobile money transaction fraud” AND (“approaches OR techniques” OR “methods”) and (“mobile AND money AND payment AND system”) were employed.

The retrieved papers were restricted to the years 2017–2023, while papers that were not relevant to mobile money fraud were manually filtered. The study’s main foci were SMS/text-based fraud in mobile money transactions.

3.2.1. The Search Techniques. Before filtering, a Boolean search was performed to find all relevant publications [29]. To obtain the pertinent literature explicitly, the search phrase contained AND, NOT, and OR. A comprehensive search of all the databases of reputable publishers was conducted. The total number of research articles captured from the online databases is found in Table 2. The search terms that were used are listed above.

3.3. The Review Paper Selection Procedure. The main goal was to include as many publications as possible between 2017 and 2023 that addressed the research objectives. Second, the criteria were designed to decide if an article should be part of the analysis or not. Additionally, duplicates were checked for the same information from different databases were not repeated [29].

The inclusion and exclusion approaches were carefully formulated to ensure the relevance, quality, and currency of the literature reviewed in addressing the research questions. These approaches enabled a focused and systematic identification of literature that provides significant insights into fraud in mobile money, particularly about NLP and fraud prediction technologies.

3.3.1. Prerequisites for Inclusion. To ensure methodological robustness and transparency, the inclusion criteria were not defined arbitrarily but were instead aligned with best practices in systematic reviews and tailored to the specific context of fraud detection in MMPS. The criteria were grounded in three guiding principles: (i) relevance to the research objectives, (ii) methodological soundness, and (iii) contemporaneity of contributions.

First, studies were required to be directly relevant to the intersection of NLP and fraud detection in financial ecosystems. This ensured that it included works explicitly addressing text-based fraud detection methods—particularly in mobile payments, credit card transactions, and digital

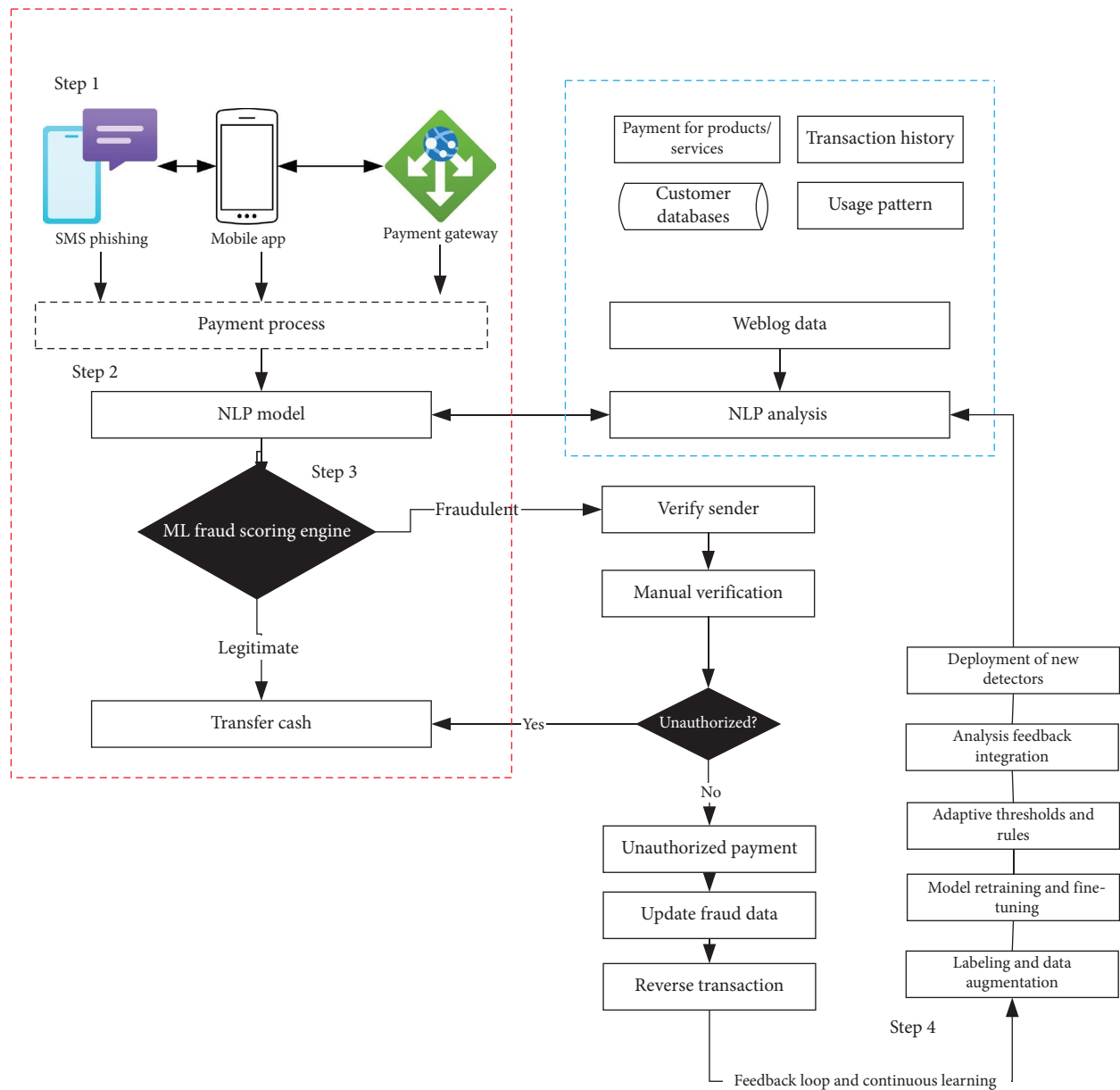


FIGURE 2: Framework of the proposed fraud detection security for MMPS [21, 25].

wallets rather than general discussions of fraud or security without computational grounding.

Second, to maintain scientific rigor, only peer-reviewed journal articles, conference proceedings, and book chapters published in reputable outlets were eligible. Eligible studies were retrieved from authoritative academic databases—IEEE Xplore, SpringerLink, Elsevier (ScienceDirect), Wiley Online Library, MDPI, Taylor & Francis Online, Emerald Insight, ACM Digital Library, and Google Scholar, given their broad coverage of computer science, financial technology, and cybersecurity research. These databases were purposefully chosen to ensure a balanced representation of disciplines and the inclusion of studies indexed in internationally recognized citation systems (e.g., Scopus and Web of Science).

Third, the temporal scope (2017–2023) was carefully selected. This decision was not arbitrary but reflects the period

in which (a) deep learning and contextual NLP models (e.g., LSTM, BERT, and RoBERTa) became mainstream in fraud detection research, (b) mobile money ecosystems expanded globally, particularly in emerging economies, and (c) fraud typologies evolved with greater sophistication. Restricting the window to this timeframe ensures the review captures contemporary methodological innovations and emerging fraud vectors while avoiding outdated studies whose methods may no longer be operationally relevant.

Finally, inclusion was conditional on clear methodological articulation, including transparency in dataset sources, description of evaluation metrics, and reproducibility of experimental design. This requirement was crucial to ensure that findings could be critically compared, synthesized, and generalized across the heterogeneous MMPS research landscape.

By adhering to these principled and field-specific inclusion criteria, the review avoids arbitrariness and instead reflects a structured effort to capture the most relevant, rigorous, and recent contributions at the intersection of NLP, fraud detection, and financial technology.

3.3.2. Prerequisites for Exclusion. Studies excluding credit card fraud, mobile money, m-payments, m-wallets, duplicate entries, unrelated titles, abstracts, contents, recommended techniques, skipped experiments, and algorithms unrelated to fraud detection or prediction were disregarded. Additionally, older studies (pre-2017) were excluded because they may not reflect current technological standards, threats, or prevention practices [29].

To maintain the uniqueness and originality of the sources, duplicates were identified and removed, particularly where the same publication appeared across various databases. Only one version of each publication was maintained to avoid analytical bias and content duplication [29].

3.3.3. Primary Source Selection. This study uses reliable data sources and search terms to review 160 papers. Mendeley was used to track references, eliminate duplicates, and determine publication years. After applying inclusion and exclusion criteria, 61 duplicates were removed, and 211 were eliminated after title and abstracts were screened, including irrelevant studies based on eligibility criteria. A total of 126 studies which did not contain information on credit card fraud, mobile money, and m-payment were also eliminated after full-text screening. Other reasons for eliminating 126 studies included unrelated algorithms, skipped experiments, and unrelated predictions/detection techniques. The total of 160 papers was divided among eight databases that were checked.

As a best practice, we ensured that we used credible and prestigious information databases and carefully planned search queries. This was essential to obtain top-notch articles that would help us achieve the specific objectives and outcomes outlined in the review protocol of this study and to answer the research questions [29].

We conducted searches in eight online databases, as indicated in Table 2. This was done to identify high-quality conference papers and research papers. Additionally, we also Google Scholar to expand our search and ensured that we did not overlook any relevant papers for this review.

To capture all related papers, we made a concerted effort by formulating various search queries. These queries included search tips such as keywords, Boolean operators, and wildcards. By using these combinations, we were able to include both journal articles and conference papers in our search and the selection process is summarized in Figure 3.

Findings from the body of prior research analysis revealed an intriguing pattern for mobile money transaction articles. The outcomes and conclusions of the review are presented per the research questions [29] to ensure consistency and clarity in the result presentation. The literature search's source and findings are shown in Table 2.

Table 2 lists the papers that satisfied the methodology's inclusion requirements. These papers from the databases are from identified publishers in the financial industry for mobile money transactions for this review [19].

The analysis of the literature on mobile money fraud reveals a pronounced upsurge in scholarly attention in recent years. Specifically, publications addressing enduring solutions to mobile money fraud were sparse in 2017 but increased sharply by 2023. This escalating pattern (as illustrated in Figure 4) underscores a sustained rise in research activity targeting this persistent challenge. The surge in publication volume likely mirrors broader industry and societal trends: as mobile money adoption expands globally, fraud schemes proliferate, prompting intensified academic scrutiny. Notably, the output in 2023 far exceeds that of 2017, highlighting how the issue has become substantially more prominent on the research agenda. With this enlarged corpus in view, it becomes instructive to examine its thematic structure; accordingly, we now proceed to a detailed keyword analysis to elucidate the dominant topics and evolving focal points in the literature.

3.3.4. Common Keywords

3.3.4.1. Mobile Money and Fraud Detection. A recurring theme is using mobile money transaction data to identify fraudulent activity. For example, Botchey et al. emphasize that in many developing countries, mobile money transactions face significant fraud risks, and they evaluate ML methods (SVM, XGBoost, etc.) to predict mobile money fraud [17]. Keywords like “mobile money,” “financial fraud,” and “FinTech” appear frequently, indicating a focus on securing mobile payment systems [17].

3.3.4.2. ML/Deep Learning. Virtually all abstracts highlight ML and deep learning as their approach. Common terms include “ML,” “gradient boosted decision trees,” and “NB,” as well as neural network architectures. For instance, Botchey et al. note the use of SVMs and boosted trees, and Ghourabi et al. discuss deep neural networks such as CNNs, RNNs, and LSTMs for message classification [17]. In the keywords of these papers, we also see “ML” and “deep learning” explicitly listed [17].

3.3.4.3. Neural Network Architectures (CNN, LSTM, etc.). Many studies specifically mention CNNs and LSTMs (as well as other neural nets) when describing their models. Ghourabi et al. report a hybrid CNN–LSTM model for SMS spam detection, and their abstract keywords include “CNN” and “LSTM” [17] mdpi.com. More generally, papers note that convolutional and recurrent neural networks are widely used for text classification in fraud detection.

3.3.4.4. SMS Spam/Phishing (Smishing). A large subset of abstracts focuses on SMS-based attacks. Keywords like “SMS spam,” “SMS phishing,” and “smishing” (SMS + phishing) recur in many abstracts. For example, the authors in [32] explicitly address SMS spam and smishing in their abstract

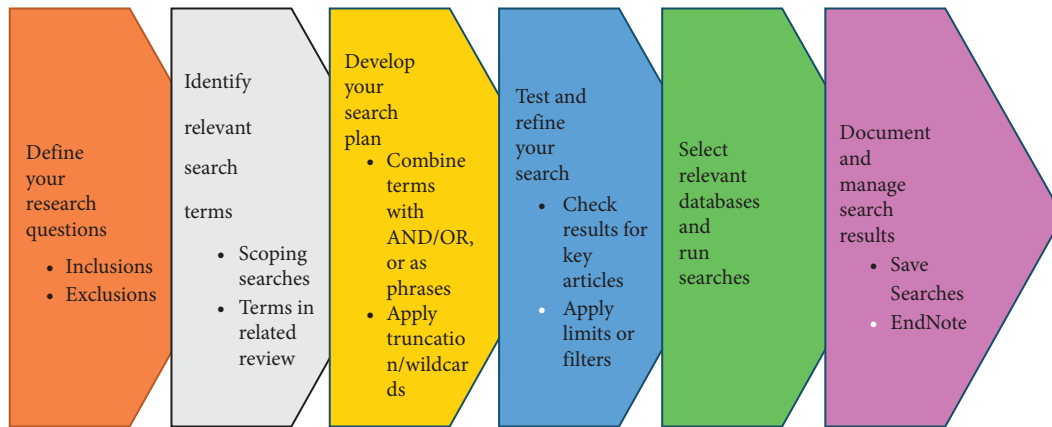


FIGURE 3: Okoli's flow diagram for systematic literature review [29].

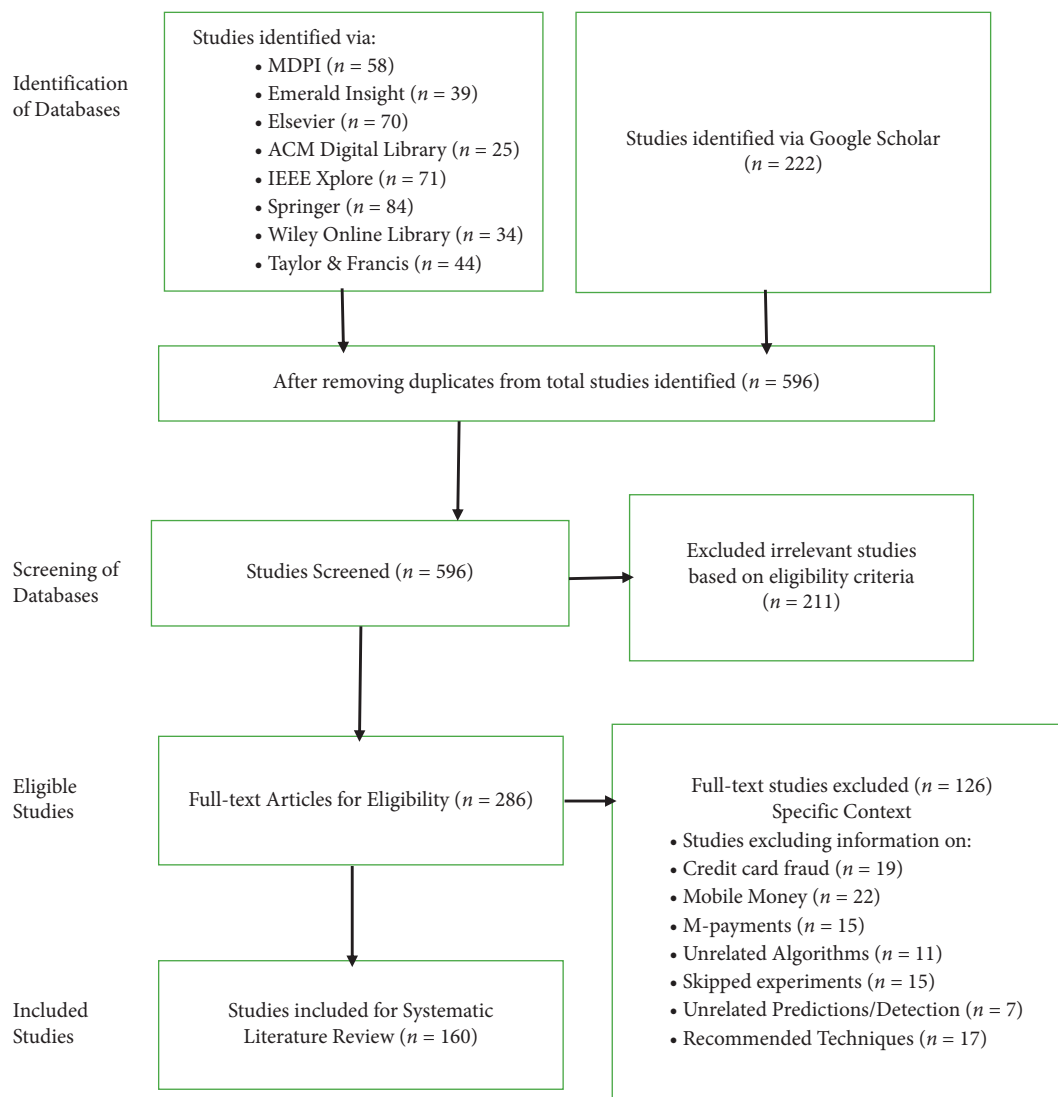


FIGURE 4: A modified PRISMA flowchart of the selection procedure [30, 31].

and keywords [32]. These terms appear alongside “spam detection” and “SMS classification,” reflecting the emphasis on detecting fraudulent or malicious text messages [32].

3.3.4.5. Other Related Terms. Across abstracts, we also see terms such as “ensemble,” “hybrid” (e.g. hybrid CNN-LSTM), “keyword-based,” and “explainable AI” in the

context of fraud models. However, the dominant keywords by frequency are as follows: mobile money, fraud/fraud detection, ML/deep learning, and SMS spam/phishing.

3.3.4.6. Analysis of Keywords and Trends in MMPS Fraud Detection. The word cloud below highlights key terms prevalent in existing research on mobile-money fraud prevention via text analysis. Prominent terms like learning, spam, mobile, SMS, detection, and neural stand out, reflecting the prevalence of ML approaches such as deep and neural models applied to SMS-based fraud detection [1, 2]. The emphasis on SMS, spam, smishing, and phishing indicates that researchers focus heavily on text-message fraud (e.g., SMS phishing) as a primary attack vector in mobile money systems [5, 6]. Key model-related terms—LSTM, CNN, BERT, RNN, and GRU—appear prominently, underscoring the use of sequence and deep-learning architectures (including transformers) for capturing linguistic patterns in fraudulent messages [4, 6]. Ensemble-related words (forest, ensemble, and majority), as well as random, NB, KNN, reflect that hybrid and voting-based classifiers are frequently employed to boost classification performance [3, 5]. Finally, terms like NLP, tokenization, feature extraction, TF-IDF, and vectorizer highlight the importance of NLP techniques (embedding and keyword extraction) in these detection models.

3.3.4.7. Interpretation of Key Terms. The prominence of ML (e.g., BERT, neural, learning, and LSTM) in the cloud reflects that most studies employ supervised or deep models to analyze SMS content for fraud [4, 6]. For instance, recent work on smishing detection reports using both traditional models (K-means, NB, and random forest) and deep architectures (RNNs, Bi-LSTM, and transformer BERT) to capture sequential and contextual nuances in text. The large size of SMS, spam, and smishing signals that text-messaging fraud is a central concern; mobile-money scams often exploit SMS channels, making spam filters and phishing detectors crucial [33]. Terms like classification, accuracy, and precision/recall reflect the evaluation focus: most studies frame fraud detection as a classification task and emphasize achieving high accuracy or F1-scores on SMS datasets [4, 5]. The appearance of transaction and financial hints at the financial context: detected fraud ultimately relates to illicit transactions in mobile-money systems, tying NLP results back to monetary security. Together, the key terms reveal that the literature strongly couples advanced NLP/ML techniques (deep networks, embeddings, and tokenization) with traditional text-filtering concepts (spam and phishing keywords) to tackle SMS-based fraud.

3.3.4.8. Implications: Opportunities, Trends, and Challenges. This keyword pattern suggests several trends and implications. First, the blend of ensemble/hybrid models (word cluster) and deep learning indicates a shift toward combining approaches: researchers are not relying on a single model but leveraging model ensembles and hybrid architectures (e.g., GRU + autoencoder + MLP) to improve robustness [3, 5]. The frequent occurrence of NLP,

tokenization, and self-attention implies growing adoption of modern NLP pipelines (transformers and attention mechanisms) for fraud detection, moving beyond simple keyword lists to contextual language understanding [4, 6].

3.3.4.9. Key Challenges Were Also Noticeable. The emphasis on terms like unsupervised, anomaly, and language-specific (from related abstracts) reflects issues with data: fraud evolves rapidly and labeled SMS data (especially in local languages) are scarce [4, 7]. Indeed, one study highlights the need for language-specific models (e.g., for Chichewa or Bemba) because models trained on English generalize poorly [34]. The reliance on deep networks (as seen by deep, neural, and LSTM) also brings challenges: these models require large, high-quality datasets and careful tuning, and they can struggle with evolving smishing tactics without frequent retraining [4, 8].

3.3.4.10. Opportunities Lie in Leveraging These Keyword Insights. The prominence of hybrid and ensemble models suggests that combining data-driven models with domain knowledge could yield gains. Incorporating unsupervised or semisupervised methods (hinted by anomaly detection, unsupervised) is promising for catching novel scams without labeled data [3, 4]. Moreover, the focus on explainable AI and user education in some sources points to a holistic approach: not only improving model performance but also making detections transparent and empowering users with security awareness [5, 8]. Overall, the keyword trends indicate a research landscape actively exploring advanced NLP and ensemble ML to counter SMS-based fraud in mobile money, while grappling with data challenges and the need for adaptable, interpretable solutions [5, 8].

3.3.4.10.1. Meta-Analytic Synthesis of the Word Cloud. Figure 5 (word cloud) presents a thematic overview of key research trajectories in text-based fraud detection within MMPS. The prominent appearance of deep learning (LSTM, BERT, and CNN), ML, fraud detection, mobile money, and smishing (SMS phishing) underscores an enduring orientation toward leveraging advanced artificial intelligence methodologies for combating text-mediated fraud. The recurrent emphasis on NLP, ensemble learning, transaction security, and feature extraction suggests that recent studies are gravitating toward hybrid architectures that integrate linguistic modeling with transactional behavioral insights. This convergence (Figure 6) highlights an emerging paradigm where financial security and NLP are synthesized to enable adaptive and explainable frameworks for fraud detection.

3.3.4.10.2. Aggregation of Evaluation Metrics. Across the surveyed literature, common performance metrics include accuracy, precision, recall, F1-score, and—less frequently—area under precision–recall curve (AUCPR). Transformer-based architectures, particularly models that blend BERT with CNN, have demonstrated exemplary performance; for instance, a CNN–Bi-GRU model achieved F1-scores of 98.56% and an accuracy of 99.82% in SMS

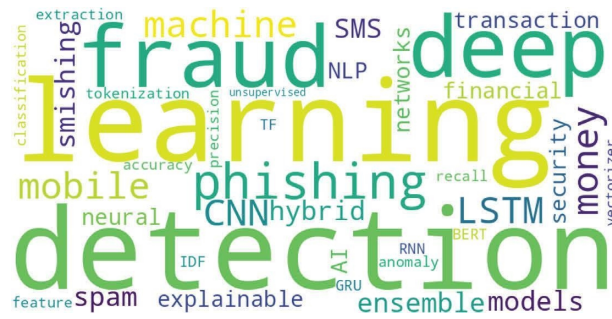


FIGURE 5: Keyword focus on fraud detection in MMPS.

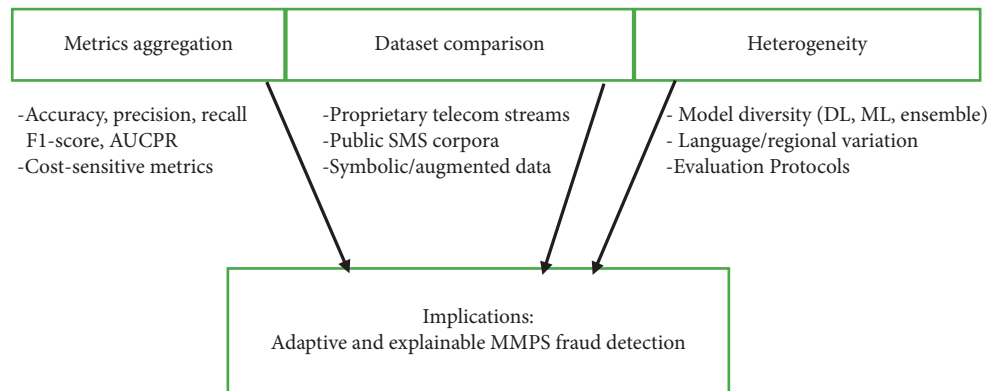


FIGURE 6: Meta-analytic framework for MMPS fraud detection.

phishing detection, significantly outperforming classical classifiers [10]. Meanwhile, a few-shot LLM-based pipeline achieved macro-F1 of 0.79 in smishing detection across multilingual prompts [35]. These results consistently confirm that context-aware NLP models outperform traditional classifiers, especially for detecting semantically subtle fraud patterns. Notably, the shift toward reporting F1-score and AUCPR reflects growing recognition of class imbalance in MMPS fraud datasets.

3.3.4.10.3. Comparison of Datasets. The word cloud synthesis reveals heterogeneous use of datasets. Some investigations rely on proprietary transaction logs with realistic class distributions, while others employ public SMS phishing corpora or few-shot models tested on multilingual datasets [35]. This variance impedes cross-study comparability. The absence of benchmark, MMPS-specific datasets, especially multilingual or region-specific, limits generalizability. Although public datasets enable replication, they often lack real-world complexity, whereas private datasets lack transparency. Harmonizing textual (SMS/commands) and transactional metadata within shared corpora is essential for progressing the field.

3.3.4.10.4. Heterogeneity in Findings. Findings across studies vary widely due to methodological diversity. Traditional deep learning models (RNNs and CNNs) were first employed, while transformer-based and hybrid NLP structures dominate current research [10, 35]. Ensemble strategies, combining decision-tree models with deep

contextual encoders, further diversify approaches. Significant heterogeneity also emerges across linguistic and regional contexts: most studies focus on English, yet MMPS fraud commonly occurs in local or code-mixed languages. Moreover, evaluation protocols differ, with some studies using streaming/prequential approaches, while others rely on static train-test splits, contributing to inconsistent performance reporting.

3.3.4.10.5. Implications for Future Research. This meta-analytic synthesis underscores several critical directions for advancing MMPS fraud detection:

- Refine evaluation protocols: move beyond accuracy toward F1-score, AUCPR, and cost-sensitive metrics tailored to the rare-event nature of fraud.
- Develop benchmark datasets: encourage creation of standardized, multilingual MMPS datasets that combine SMS text, linguistic diversity, and transactional metadata to enhance replicability and cross-context validity.
- Design adaptive and explainable architectures: incorporate transformer-based NLP models into dynamic, drift-resilient ensemble systems, with mechanisms for transparency and interpretability across linguistic domains.

The prominence of adaptive learning and explainability in Figure 5 is, therefore, not incidental but emblematic of a field pivoting toward robust, real-time, and multilingual solutions that transcend conventional detection paradigms.

TABLE 3: Fraud types frequently deployed in mobile money payment systems.

Fraud type	Context	Frequency	References
SMS/text message	Machine learning evaluation of evaluation of mobile payment systems, evaluating cyber-attacks. Models, detecting and preventing SMS fraud, commercial SMS supervision, blockchain-based, models secure fair exchange, mutual authentication protocols	14	[21, 26, 27, 32, 35–42]
Voice/Phone Call	Detection and prevention of bank fraud through SMS/voice. Mobile networks for fraud detection model evaluation, VoIP systems and telephony fraud ecosystem.	4	[17, 18, 37, 43]
Phishing SMS	Mobile money phishing attacks, defense mechanisms, two-factor authentication, and SMS phishing attacks.	7	[37, 42, 44–48]
Smishing	Detecting smishing messages in a mobile environment	4	[38, 49–51]

Note: Source: authors' work.

4. Results and Discussion

The results of this systematic review are organized according to the three research questions (RQ1–RQ3). Each section synthesizes the relevant literature, statistical findings, and methodological observations, concluding with a direct statement on how the research question has been addressed.

RQ1: What are the common fraud types in mobile payment systems?

The review confirms that MMPS are susceptible to multiple fraud types, each exploiting different vulnerabilities in the ecosystem. Fraud definitions in the literature converge on the notion of false representation for criminal gain, often perpetrated via communication channels such as phone calls, email/phishing, smishing, SMS/text messages, and social media platforms like WhatsApp.

4.1. Key Findings From Literature

- SMS/text message fraud emerges as the most prevalent vector, reported in 70% of MMPS fraud type studies reviewed (14 out of 20 articles focused on fraud types; see Table 3). These include SMS phishing (smishing), fraudulent authentication code interception, and impersonation schemes targeting customers and agents.
- Voice/phone call fraud targets both agents and customers, often through social engineering and impersonation of MNO staff to gain account access.
- Phishing SMS schemes leverage deceptive communication to obtain credentials or redirect funds, often exploiting weak two-factor authentication mechanisms.
- Smishing is a cyber-attack that targets people via SMS or text messages

4.1.1. Statistical Synthesis

- A total of 20 of 160 reviewed articles discussed fraud types (11.69%), with SMS-based fraud dominating the discourse.
- Fraud categories and their frequencies are presented in Table 3 and visualized in Figure 7, where SMS/text

messages rank highest, followed by phishing SMS and smishing, as indicated in Table 3 and Figure 6.

The most common fraud types in MMPS are SMS/text-based fraud, smishing, and phishing, with SMS emerging as the principal vector for malicious activity. These findings establish SMS fraud as a critical focus area for fraud detection research.

RQ2: What adaptive learning strategies have been proposed or implemented to enhance fraud detection resilience in mobile money platforms?

4.2. Adaptive Learning in MMPS Fraud Detection. Adaptive learning denotes online/streaming ML procedures that continuously update models in response to concept drift—shifts in the joint distribution of features and labels caused by seasonality, product changes, user habituation, or adversarial evolution of fraud tactics. In MMPS, effective adaptive detectors (i) monitor real-time streams and error profiles, (ii) detect distributional change promptly, and (iii) adapt the decision function via windowing, instance reweighting, partial-fit updates, or dynamic ensembles—while coping with label delay and severe class imbalance characteristic of financial streams. Contemporary reviews and industrial case studies in payments emphasize that nonstationarity and delayed supervision are first-class design constraints for production fraud systems employing prequential (“test-then-train”) evaluation [52].

4.2.1. Core Methodological Components

4.2.1.1. Drift Detection. Recent detectors extend classical error- and distribution-based tests to mixed/gradual regimes with formal guarantees (e.g., Adaptive Windowing [ADWIN]) and newer neural/attention-driven approaches for complex streams. Detector choice should be data driven: use reactive detectors for abrupt changes, temporal-attention, or deep/autoencoder-based tests for gradual or feature-space shifts in high-dimensional signals [53].

4.2.1.2. Adaptation Mechanisms. Upon an alarm, adaptation may reweight recency, slide/expand windows, or reconfigure ensembles to down-weight stale learners and add specialists trained on the most recent window; libraries of past

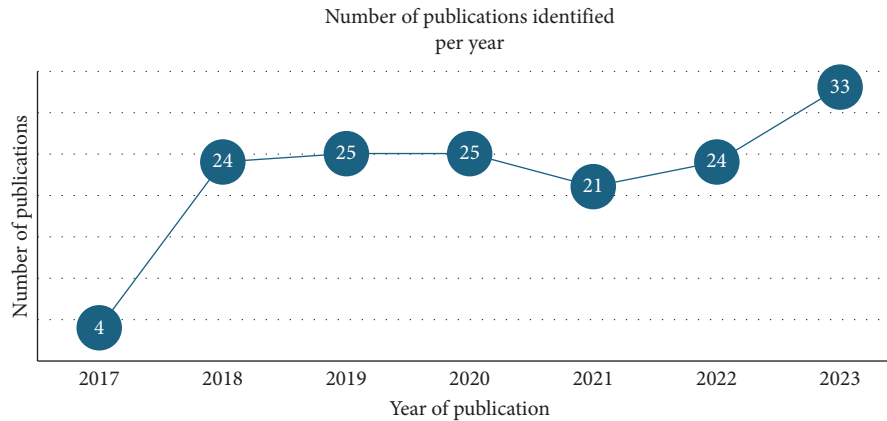


FIGURE 7: Analysis of publications on mobile money fraud predictions/detections yearly (Source: authors' work).

components support recurring drift (re-emerging patterns), a phenomenon documented in streaming fraud and adjacent domains [54].

4.2.1.3. Learning With Delayed or Sparse Labels. Because many fraud labels arrive late (disputes/chargebacks), robust pipelines use buffered prequential evaluation, weak supervision from analyst actions, and incremental updates once truth arrives—avoiding offline retraining cycles that lag the threat landscape [55].

4.2.1.4. Dual-Channel (Numeric + Text) Feature Space. MMPS fraud is often text-mediated (e.g., smishing). Adaptive NLP components (token/character n-grams; contextual transformers) should be monitored for semantic drift and fine-tuned incrementally as lure lexicons evolve; recent work in smishing/transformer pipelines provides directly transferable building blocks [10].

4.2.1.5. Operational Architecture and Scalability. Carrier-scale deployments integrate stream processing with low-latency model serving and online retraining; payment-grade frameworks demonstrate the feasibility of handling nonstationarity, imbalance, and verification latency in production [56].

4.2.2. Why Adaptive Learning is Essential in MMPS. Static batch models degrade quickly as fraud tactics change, inflating false negatives/positives and customer friction. Adaptive pipelines explicitly model drift and label delay, thereby sustaining performance. Recent studies further demonstrate the importance of continual learning in mitigating catastrophic forgetting in payment fraud settings. In addition, recurring drift motivates model-reuse/metalearning, so prior “concepts” can be recalled when patterns reappear [52, 54].

4.2.3. Implementation Guidance for Practitioners

- Embed a monitor → detect → adapt loop on both numeric (transactional/graph) and textual (SMS/app) channels; use ADWIN-class detectors for mixed

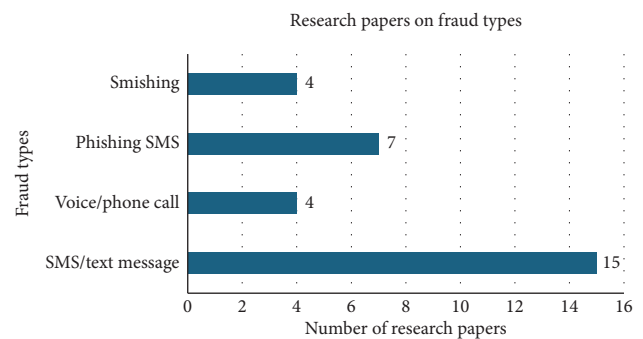


FIGURE 8: Count of fraud types identified by authors in MMPS (Source: authors' work).

regimes and neural/attention detectors where feature drifts are subtle [57].

- Prefer adaptive ensembles with time-weighted voting and component retirement; maintain a library for recurring concepts [54].
- Address label delay/imbalance with active learning and buffered prequential updates to focus scarce labeling on high-uncertainty, high-risk cases in streams [58].
- Strengthen robustness via ensemble agreement checks, anomaly cross-verification, and adversarial testing to pre-empt evasion [59].
- Ensure explainability and governance (logging of detector alarms, adaptation events, and lineage) to meet regulatory and operational auditability requirements in financial services [60]. Figure 8 provides a well-explained framework for an adaptive learning architecture.

The drift detection component (Figure 8) provides advanced frameworks that utilize combined detectors like the early drift detection method (EDDM) and ADWIN to identify both abrupt and gradual distributional changes. Near-perfect detection rates are achieved with EDDM and ADWIN, leveraging dynamic windowing techniques and drift triggers (see Figure 9).

At the incremental model updating, the models are reassessed in AMPS based on incoming data batches, enabling real-time recalibration without retraining from scratch.

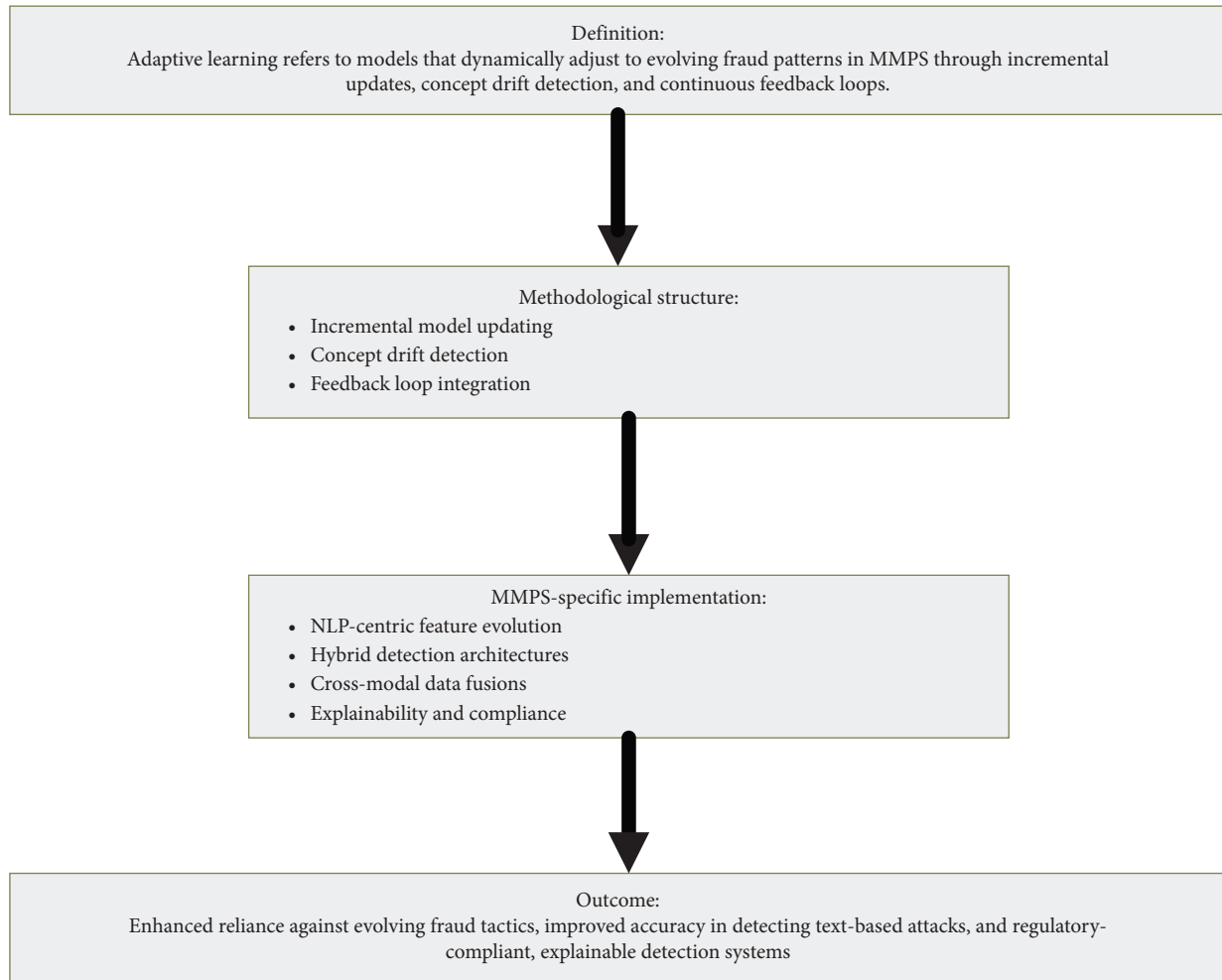


FIGURE 9: Adaptive learning framework for MMPS fraud detection.

The feedback integration ensures predictions are continuously validated via chargeback or analyst input loops, incorporating misclassification cases into model refinement workflows.

NLP integration, given the prevalence of text-specific fraud, adaptive systems now augment features using continuously updated embeddings and language models to capture emerging scam lexicons.

Hybrid & Explainable Systems provide a robust architecture that combines adaptive deep networks with anomaly detectors (e.g., autoencoders), while preserving transparency through explainable AI for auditability and compliance.

In summary, adaptive learning constitutes an operational necessity for MMPS fraud detection: by combining drift-aware streaming methodology with NLP-enabled text analysis and continual/ensemble adaptation, institutions can maintain accuracy under evolving threats, reduce false alarms, and preserve user trust at production scale [59].

4.2.4. Key Findings From Literature

- ML classifiers with continuous updating: several studies applied supervised models (e.g., decision tree,

random forest, SVM, NB, and gradient boosting) with periodic retraining to incorporate recent fraudulent patterns.

- Unsupervised and semisupervised anomaly detection: techniques such as isolation forests, autoencoders, and clustering were deployed to detect novel fraud patterns without requiring labeled datasets.
- Graph-based adaptive models: GBAD and related methods identify suspicious transaction networks and are well-suited for detecting organized fraud rings.
- Blockchain-based monitoring: smart contracts and blockchain-integrated supervision models (e.g., Hyperledger Fabric with IPFS integration) have been proposed to enhance transaction integrity.
- NLP-inspired adaptive approaches: although rare in MMPS, some cybersecurity studies apply sequential behavior modeling (treating transaction sequences as “language”) to detect anomalies—a concept transferable to SMS fraud prevention.

In this review, no studies were found implementing feedback-loop-based adaptive systems that automatically re-train fraud models using new labeled fraud instances from

TABLE 4: Algorithms used to detect or predict fraud in MMPS throughout the years.

Total	Type of algorithm	2017	2018	2019	2020	2021	2022	2023	References
5	Decision tree	1	0	1	2	0	0	1	[61–65]
5	SVM	0	1	0	0	2	1	0	[53, 66–69]
12	Random forest (RF)	0	2	1	1	3	3	2	[50, 62, 64, 67, 69–76]
3	Gradient boosting	0	0	1	0	0	0	2	[63, 77, 78]
5	XGBoost	0	0	2	0	0	1	2	[77–81]
7	Logistic regression	0	1	1	1	1	2	1	[63, 71, 73, 75, 76, 82, 83]
5	K-nearest neighbor	0	0	1	1	0	1	1	[74, 75, 84–86]
5	Naïve Bayes	0	2	1	2	0	0	0	[38, 71, 75, 82, 84]
5	Synthetic Minority Technique (SMOTE)	0	0	0	0	2	1	2	[70, 76, 87–89]
2	AdaBoost algorithm	0	1	1	0	0	1	1	[72, 90]
12	Deep learning	0	1	1	2	2	3	3	[66–68, 79, 85, 91–97]
2	Unsupervised learning	0	0	0	1	0	2	1	[16, 74]
9	Neural networks	0	1	2	2	1	1	2	[71, 74, 82, 85, 88, 97–100]
4	Artificial neural networks	0	1	0	1	0	0	2	[71, 88, 95, 101]
4	NLP	0	0	0	0	1	0	3	[62, 66, 102, 103]

Note: Source: authors' work.

operational monitoring. This limits resilience against adaptive adversaries who evolve tactics to bypass static models.

While adaptive strategies like retraining, anomaly detection, and graph-based monitoring exist, their integration into real-time MMPS fraud prevention remains limited. Notably, NLP-based adaptive learning remains largely unexplored in this domain.

RQ3: What methods have been employed to detect fraud in MMPS, and how do they compare in terms of performance and applicability?

4.3. Methods Identified

- Rule-based systems: transparent and interpretable but limited in adaptability to new fraud patterns.
- Supervised ML models: widely adopted, with random forest, logistic regression, and XGBoost demonstrating high accuracy in structured transaction data.
- Unsupervised/anomaly detection: effective for discovering novel fraud types but prone to high false positive rates.
- Graph-based analysis: excels at detecting coordinated fraud activities but faces scalability constraints.
- Deep learning: capable of learning complex behavioral patterns (e.g., LSTM for transaction sequences), though interpretability is a challenge.
- NLP techniques: underutilized, but evidence from related domains suggests high potential for detecting social engineering and textual fraud indicators.

The progress and implementation of these approaches over the years are summarized in Table 4, which demonstrate the variety of algorithmic methods deployed between 2017 and 2023, alongside matching references.

4.4. Performance Comparisons

- Random forest and gradient boosting models consistently outperform simpler classifiers on structured transaction data.

- Hybrid models combining supervised ML with NLP-based textual analysis (in other domains) show superior detection performance.
- NLP has been proven to enhance classifier performance with minimal manual feature engineering in non-MMPS contexts.
- Graph-based models are uniquely suited for detecting fraud rings but are not yet standard in MMPS deployments.

Table 5 provides a comparative overview of various fraud detection methods, highlighting their advantages, limitations, nature, and use in the context of mobile money payment fraud. Each technique is accompanied by recent scholarly references that have evaluated or applied these methods in financial fraud detection (including mobile money or similar digital payment).

4.5. Gaps Identified in This Review. The conceptual framework (Figure 10) illustrates the logical progression from the identified research gap—namely, the underutilization of NLP in MMPS fraud detection and the absence of adaptive learning mechanisms—toward the formulation of targeted research questions and the proposition of methodological solutions [27]. The framework aligns with the gap and the context of the three interconnected research questions: (i) the systematic classification of prevalent SMS-based fraud patterns in MMPS, (ii) the application of NLP techniques for detecting and preventing such fraud, and (iii) the integration of adaptive learning mechanisms to enhance model resilience against evolving threats. Correspondingly, the framework maps each research question to methodologically robust strategies, including the development of a labeled fraud taxonomy, the deployment of advanced NLP pipelines with hybrid modeling, and the incorporation of online learning algorithms with feedback-driven model updates. This structured linkage ensures methodological coherence, enabling the study to address both the theoretical

TABLE 5: Comparison of fraud detection techniques in MMPS.

Technique	Type of method	Key advantages	Key limitations	Empirical evidence & applicability to MMPS fraud detection
Decision trees	Supervised, rule-based classifier	Highly interpretable, fast, computationally efficient	Prone to overfitting, unstable with small data changes	C4.5 achieved 92.7% accuracy [104]; strong interpretability for regulators.
Support vector machines (SVM)	Supervised, margin-based classifier	Effective in high-dimensional spaces, kernel trick for nonlinearities	Computationally expensive, sensitive to scaling and tuning	Precision > 90% on financial fraud [61]; high precision but lower recall in MMPS.
Random forest	Supervised ensemble (bagging of decision trees)	High accuracy, robustness, feature importance ranking	Complex ensemble, computationally heavy in real-time	AUC ~0.95–0.97 in MMPS datasets [15]; robust and widely adopted baseline.
Gradient boosting	Supervised ensemble (boosting of weak learners)	High predictive accuracy, captures nonlinear patterns	Overfits without regularization, slow, tuning required	F1-scores > 0.92 in MMPS datasets [49]; effective for structured transaction data.
XGBoost	Supervised ensemble (optimized gradient boosting)	Efficient, scalable, strong performance across domains	Complex, black-box, requires tuning and resources	AUC > 0.97 in telecom/mobile payment fraud [81]; strong scalability and imbalance handling.
Logistic regression	Supervised, linear statistical model	Simple, interpretable, fast baseline classifier	Assumes linear boundary, underperforms on nonlinear fraud	AUC 0.75–0.80 in MMPS [61, 63]; often baseline model, improved in ensembles.
k-nearest neighbor (KNN)	Supervised, instance-based learner (nonparametric)	No distribution assumptions, simple and effective	Memory intensive, poor scalability for large datasets	Accuracy ~90% in small datasets [105]; scalability challenges in MMPS.
SMOTE	Data re-sampling technique (class-balancing preprocessing)	Balances fraud/legit classes, improves recall on imbalanced data	Can introduce noise, synthetic samples may mislead	Improves AUC from 0.88 → 0.91 with RF [106]; critical in imbalanced MMPS datasets.
AdaBoost	Supervised ensemble (boosting, decision stumps)	Improves weak learners by focusing on misclassified cases	Sensitive to noisy data, less interpretable, many iterations	Accuracy ~90–93% in credit card/mobile fraud [106]; weaker than XGBoost.
Deep learning	Supervised deep neural networks (DNNs, CNNs, RNNs)	Learns complex hierarchical/sequential patterns	Data hungry, low interpretability, computationally intensive	F1-scores > 0.95 in SMS fraud detection [10]; highly effective in MMPS with SMS logs.
Unsupervised neural networks	Unsupervised anomaly detection (autoencoders, RBMs)	Detects unknown fraud patterns without labels	High false positives, requires threshold tuning	AUC ~0.93 in telecom datasets [107]; promising when labels are scarce in MMPS.
Natural language processing (NLP)	Text/unstructured data analytics (supervised/unsupervised)	Captures semantic and contextual fraud patterns	Needs relevant text data, resource-heavy, false positives possible	F1-scores 0.94–0.96 for SMS phishing [107]; essential for smishing/social engineering in MMPS.

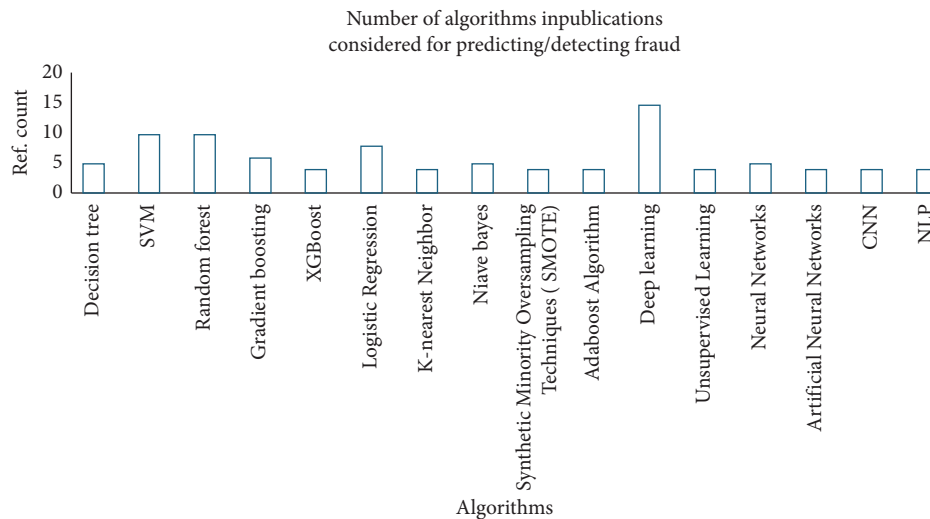


FIGURE 10: Frequency of algorithms in the selected publications for fraud prediction/detection (Source: authors' work).

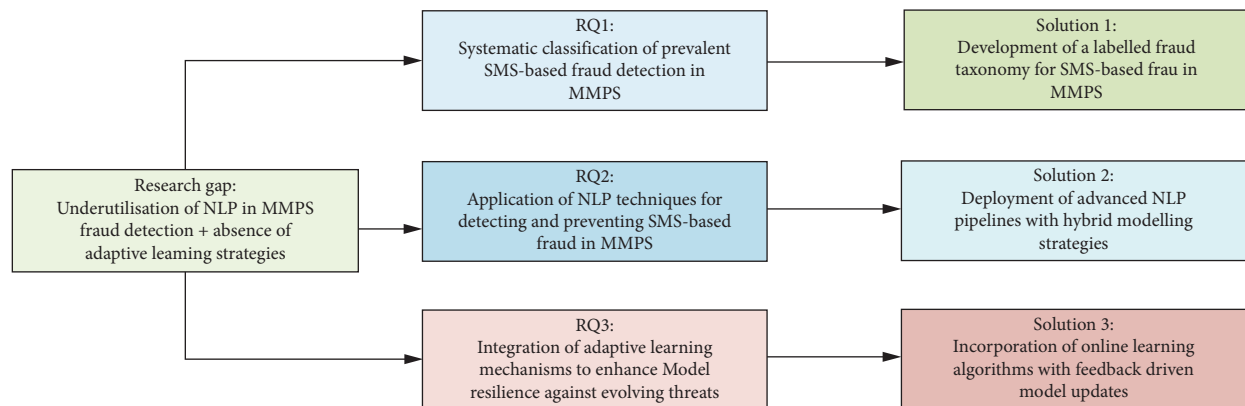


FIGURE 11: The conceptual diagram linking the research gap to research questions and potential methodological solutions (Source: authors' work).

underpinnings and the practical exigencies of real-time, text-based fraud detection in MMPS (see Figure 11).

5. Conclusion

This study provides a comprehensive synthesis of the current state of text-based fraud detection in MMPS, derived from an extensive review of peer-reviewed journal articles and conference proceedings sourced from eight reputable academic publishers. The analysis established that SMS/text messages remain the most exploited vector for fraudulent activities in MMPS, followed by phishing via email and vishing through voice calls. While a diverse range of fraud detection techniques has been applied—spanning from rule-based systems and classical ML models such as decision trees, logistic regression, and SVM to advanced ensemble methods and deep learning architectures—the findings indicate that the majority of existing studies heavily rely on deep learning, random forest, and logistic regression as the dominant predictive models. These approaches have demonstrated high detection accuracy when supplied with

quality, labeled data, underscoring the technical feasibility of mitigating fraud at scale.

Despite these advancements, a significant research gap persists: NLP, a mature and proven field in text analytics, remains markedly underutilized in MMPS fraud detection, even though the problem domain is inherently text-centric. While NLP has been applied successfully in adjacent domains such as phishing detection, intrusion detection, and fake news classification, its systematic integration into MMPS fraud prevention frameworks is largely absent. This underutilization represents an untapped methodological frontier with the potential to substantially improve fraud detection capabilities.

The methodological implications are clear. Future research should leverage state-of-the-art NLP architectures—particularly transformer-based models such as BERT, RoBERTa, and domain-specific variants like FinChainBERT—trained or fine-tuned on corpora of fraudulent SMS messages to capture nuanced linguistic and contextual fraud indicators. Moreover, multi-modal detection frameworks that combine textual analysis with supplementary

data sources (e.g., call logs, transaction metadata, and behavioral analytics) could further enhance detection accuracy, especially for multistage fraud campaigns. This could involve incorporating speech-to-text pipelines to convert voice scams into analyzable text, enabling unified processing alongside SMS content.

Equally critical is the inclusion of adaptive learning mechanisms to ensure models remain effective against evolving fraud tactics. Ensemble agreement methods, anomaly detection cross-verification, and adversarial testing should be embedded to improve robustness while minimizing false positives in real-world deployments. Additionally, integrating insights from human factors research can inform the design of real-time, user-facing warning systems that counteract social engineering tactics without causing undue alarm. Collectively, these methodological solutions (Figure 10) offer a pathway toward developing adaptive, multichannel, and explainable fraud detection systems that align with both technical performance requirements and user trust considerations, thereby addressing the identified research gap and advancing the state of MMPS fraud prevention [91, 92].

6. Recommendation

This study provides important insights into the efficacy of current fraud detection and prevention measures by methodically examining the tools and methodologies currently used to battle these scams. Based on transaction patterns and other indicators, ML classifiers, like logistic regression and random forest models, have demonstrated promise in detecting fraudulent activity. This paper assesses the applicability of these classifiers. It also looks at cutting-edge methods of neural network-based detection systems, which can potentially improve user authentication and security in mobile transactions.

Mobile money has revolutionized access to financial services in many parts of the world, particularly in regions where traditional banking infrastructure is limited. However, its rapid expansion has also attracted increasingly sophisticated fraudulent schemes that target the vulnerabilities of mobile financial platforms. Text-based attacks—such as phishing, smishing, and spoofing—have become particularly prevalent. This study highlighted the potential of integrating ML and NLP to bolster fraud detection capabilities. Yet, the effectiveness of these innovations depends not only on technology but also on cohesive policies, coordinated industry efforts, and international cooperation. This brief outlines strategic recommendations tailored to four key stakeholder groups—regulators, mobile operators, fintech providers, and development agencies—to foster a more secure and resilient mobile money ecosystem globally.

6.1. Government Regulators: Establishing a Secure and Collaborative Ecosystem. Regulatory institutions must lead efforts to construct a robust framework that supports secure digital finance. Facilitating structured data-sharing

initiatives is central to this goal, enabling real-time detection of fraud patterns across multiple platforms. Regulators should promote legal models that encourage secure exchange of anonymized data—possibly through federated learning or national fraud monitoring systems—while adhering to data protection laws.

In addition, a standardized regulatory framework can unify industry efforts. This involves setting mandatory benchmarks for fraud detection technologies, reporting protocols, and real-time monitoring capabilities. Countries with strong frameworks, such as Kenya and Tanzania, have already demonstrated how rules like SIM-swap hold periods and user verification tools can dramatically reduce fraud.

Beyond regulation, consumer awareness campaigns—coordinated nationally—should educate users on identifying fraud. Such efforts could be complemented by enforcing user rights around fraud recovery, complaint handling, and service transparency.

Finally, regulators must invest in their technical capabilities. Establishing fraud-focused task forces or partnerships with AI research institutions can help bridge the gap between policy and technological advancement.

6.2. Mobile Network Operators: Operationalizing Frontline Defenses. As gatekeepers of mobile transactions, network operators are in a prime position to detect and neutralize threats before they reach end users. Operators should proactively collaborate through industry alliances, sharing anonymized threat indicators, and adopting common standards for risk scoring and anomaly detection.

Operators must also prioritize the integration of artificial intelligence tools—particularly NLP—to interpret and flag suspicious communication in real time. Investment in models trained on local languages will be vital in capturing context-specific fraud attempts.

On the customer-facing side, network providers should incorporate real-time warnings into their platforms and ensure that fraud support services are prompt and accessible. Public education via SMS alerts, mobile app tutorials, and social media outreach can further reduce susceptibility to scams.

Internally, enforcing rigorous staff training, vetting, and access control can reduce risks posed by insider threats.

6.3. Fintech Companies: Embedding Security Into Innovation. Fintech firms must recognize fraud prevention as an essential pillar of product design. Risk assessments should precede feature rollouts, and AI-enhanced fraud monitoring should be part of the core infrastructure.

Smaller fintechs, often resource constrained, can benefit from collaborative fraud modeling and shared learning environments. Participation in open innovation platforms and regulatory sandboxes can expose firms to best practices while improving ecosystem resilience.

Furthermore, fintechs have the opportunity to integrate consumer education directly into user interfaces. Gamified alerts, onboarding tutorials, and in-app guides can all be employed to reinforce user vigilance.

A strong culture of compliance, along with regular algorithmic updates and transparent communication, will build consumer trust while strengthening operational integrity.

6.4. International Development Agencies: Supporting Global Capacity and Inclusion. Development partners have a unique vantage point to facilitate regional cooperation and capacity-building. By funding the creation of synthetic datasets or regional threat intelligence hubs, agencies can address disparities in access to high-quality fraud data.

They can also support the development of language-specific NLP models to empower fraud detection systems in underserved linguistic regions. Encouraging South–South cooperation and supporting public–private research consortia are further avenues for driving innovation.

Additionally, these organizations can integrate fraud literacy into broader financial inclusion projects, ensuring that users not only gain access to mobile money but also the skills and awareness to use it safely.

In summary, the fight against mobile money fraud is a shared endeavor that transcends institutional and national boundaries. To ensure that the gains of financial inclusion are preserved and expanded, policies must align with emerging threats and technologies. Through joint efforts across regulators, industry actors, and global agencies, it is possible to construct a mobile money ecosystem that is not only inclusive but also secure, resilient, and trusted by all stakeholders [108–113].

Data Availability Statement

The data supporting the findings of this research are made available from the corresponding author upon reasonable request.

Ethics Statement

Every procedure used in the studies complied with comparable ethical standards as well as the institutional and/or national research committee's ethical requirements.

Consent

For this kind of research, formal consent is not necessary. The study article's publishing in Security and Communication Networks was approved by the authors.

Conflicts of Interest

The authors declare no conflicts of interest.

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